AI-HealthGuard

AI-Based Ischemic Heart Disease Risk Prediction & Prevention System

Project Report submitted in partial fulfillment of the requirement for the degree of

Bachelor of Technology

in

Computer Science and Engineering

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DECLARATION

This is to certify that the Bachelors of Technology project entitled "AI-HealthGuard: An AI-based ischemic heart disease risk prediction and prevention system" has been completed in partial fulfillment of the requirements for the degree of Bachelors of Technology in Computer Science and Engineering at the Department of Computer Science and Engineering of GL Bajaj Group of Institutions, NH-2, Mathura-Delhi Road, P.O. Akbarpur, Mathura-281001(U.P.). The project was executed between July 2025 and March 2026 under the direction of Er. Tanya Shrivastava, Assistant Professor.

The project report is the original record of the author, and has not been submitted as a basis for another degree at a different institution. The project report is free from any form of plagiarism.

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CERTIFICATE

This is to declare that Mr. Tarun Kumar (Roll No. 2205111530052), Ms. Sakshi Rajput (Roll No. 2205111530046) and Mr. Prashant Prajapati (Roll No. 2205111530036), have contributed to the development of a technical solution called "AI-HealthGuard", to assist in evaluating risk factors and preventing ischemic heart disease. In partial fulfillment of the requirements for the award of a Bachelor of Technology degree (Computer Science and Engineering) they are providing you with this report as documentation for their original work. To my knowledge this is an original project, has not been submitted for any other degree and is eligible under the terms of copyright law.

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ABSTRACT

Ischemic Heart Disease (IHD) is a worldwide epidemic responsible for an estimated nine million deaths globally, making it the most common cause of death globally, accounting for approximately 13% of overall global mortality. IHD can be prevented by early identification of risk factors and providing appropriate lifestyle interventions. Diagnostic techniques used today, such as cardiac catheterization and angiography, prove to be costly, invasive, and too far out of reach for many families and individuals who are middle-class and/or reside in rural areas.

Furthermore, current clinical risk calculators such as GRACE and Framingham rely on linear statistical data models that do not adequately represent the complex, non-linear interactions between risk factors nor do they present any explanations of how they came to the prediction contained within their output. The goal of this project is to create an AI-driven web application called AI-HealthGuard that will allow patients to predict their own risk of developing IHD and to receive recommendations regarding preventive health care at no cost to the patient or user of the system.

Specifically, AI - HealthGuard will use 13 different routine, clinically-relevant parameter measurements that have previously been established from the UCI Cleveland Heart Disease Dataset database. Five different types of classification models (Logistic Regression, Random Forest, XGBoost, Support Vector Machine (SVM) and Neural Network) will be used to predict an individual's IHD risk as binary data.

The classification model that was selected as the primary classifier and the model that will be promoted throughout this project is the XGBoost, with a target accuracy of 90% or greater and AUC of 0.90 or greater. All models will be optimized for performance using GridSearchCV, with a 5-fold cross-validation and trained on an aggregated data set containing imputed data, OneHotEncoding, StandardScaler normalization of the dataset and SMOTE oversampling. The provision of an Explainable AI feature is also a major contribution of AI-HealthGuard.

The Explainable AI feature will use SHAP (Shapley Additive Explanations) to generate per-individual model output prediction feature importance visualizations and non-technical, plain English explanatory content for each individual, based upon their unique set of data entry points. Furthermore, an individual's Composite IHD Risk Score (0-100) will be calculated and subsequently used to generate a personalized prevention report detailing prevention actions in four categories (diet, exercise/lifestyle, referrals, etc.) that can be exported to a PDF for download, which will be presented to the user through the web dashboard generated by Streamlit.

Keywords: Ischemic Heart Disease, Machine Learning, XGBoost, SHAP, Explainable AI, Streamlit, UCI Cleveland Dataset, Preventive Healthcare, Risk Prediction.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
AUC	Area Under the ROC Curve
CAD	Coronary Artery Disease
CVD	Cardiovascular Disease
DCA	Decision Curve Analysis
ECG	Electrocardiogram
EDA	Exploratory Data Analysis
EHR	Electronic Health Records
GPU	Graphics Processing Unit
IHD	Ischemic Heart Disease
LR	Logistic Regression
MI	Mutual Information / Myocardial Infarction (context-dependent)
ML	Machine Learning
NN	Neural Network
NCD	Non-Communicable Disease
PDF	Portable Document Format
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PRD	Product Requirements Document
RF	Random Forest
ROC	Receiver Operating Characteristic
SHAP	Shapley Additive Explanations
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
TRD	Technical Requirements Document
UCI	University of California, Irvine
WHO	World Health Organization
XAI	Explainable Artificial Intelligence
XGB	XGBoost (Extreme Gradient Boosting)

CHAPTER 1
INTRODUCTION

1.1. Introduction to the Project

The leading cause of death in the world is cardiovascular disease (CVD). According to statistics reported by the World Health Organization, in 2019 CVD led to 17.9 million deaths globally - making up 32% of total deaths worldwide. Of the many sub-types of CVD - Ischaemic Heart Disease (IHD or Coronary Artery Disease (CAD) or Coronary Heart Disease (CHD) - causes more than 9 million deaths each year, accounting for nearly 13% of global mortality from all causes.

IHD occurs when the coronary arteries supply blood to the heart muscle, which gets slowly smaller because plaque builds up in them. This change usually takes a long time to happen, sometimes between 20 and 30 years; and people will usually not know there is a problem until something happens where they are unable to get blood to their hearts. Some examples are unstable angina (chest pain); acute myocardial infarction (heart attack), or suddenly dying of heart disease. Because most people do not know something is wrong until very late in the development of IHD, they have a tremendous risk of getting much worse damage to their hearts before they find out that there is an issue.

Ischaemic heart disease (IHD) has had a significant impact on low- and middle-income countries (LMICs), where more than 80 per cent of deaths globally from cardiovascular disease occur. Data suggest that in India, more than 54 million people currently have varying forms of coronary artery disease (CAD). The rapid epidemiological transition seen in India is being driven by urbanisation; physically inactive lifestyles; changes to caloric dense diets; the increasing number of people living with type 2 diabetes and hypertension, as well as high levels of psychosocial distress. There is a consensus among researchers that unless large-scale, effective prevention of CAD can be put in place in India, the proportion of the global IHD burden that will be accounted for by India will continue to grow.

The public health crisis will be made worse by the economy of Interventional Heart Disease (IHD). Heart-Related Treatments such as Coronary Angioplasty, and CABG and long-term medications, as well as cardiac rehabilitation impose a major financial strain on all patients receiving treatment for IHD. In India, many people do not have Medical Insurance, with out-of-pocket healthcare cost accounting for an even larger share of family income than healthcare costs for those with Medical Insurance, will result in the majority of people in becoming "medically impoverished" as well as catastrophic financial hardship due to the high costs of treating heart diseases and illnesses resulting from IHD.

1.2. Limitations of Existing Diagnostic and Risk Assessment Approaches

Current methods for identifying as well as predicting the risk of IHD are quite limited in terms of access, interpretability and whether or not they are population-specific. Invasive Diagnostic Tests: Although coronary angiography and cardiac catheterization are considered the gold standard for diagnosing IHD, both require a specialized catheterization lab, come with a risk of procedure complications (e.g., arrhythmia, bleeding, contrast-induced nephropathy) and cost between INR 30,000 – INR 1,50,000. Thus keeping these interventions out of reach for most middle to lower class patients, especially in rural areas. Non-Invasive Diagnostic Tests: ECG, stress echocardiography, nuclear MPS and CT coronary angiography have lower risk for procedure complications, but still require significant infrastructure (including trained cardiologists), trained support staff, and substantial financial resources that are not readily available in rural or semi-urban areas. Traditional Risk Score Systems: Framingham Risk Score (developed using a primarily white North American cohort of males in the 1960s), GRACE score, and TIMI score are all based on linear statistical models, and do not account for non-linear relationships among factors such as age, hypertension, diabetes, dyslipidemia, and smoking. They were also never validated on South Asian patients who present with IHD at a younger age and with a different set of risk factors than those used to calculate these scores. Explainability: The existing systems provide only a single risk prediction value and do so without providing any guidance about how the individual factors contribute to that prediction, making it difficult for both the clinician and the patient to

understand the basis of the assessment. There are no Actionable Preventive Recommendations: The existing risk scores do not provide patients with individualized, actionable preventive recommendations which can be performed immediately.

1.3. Machine Learning and Explainable AI as Transformative Technologies

Machine Learning (ML) has a more advanced way to predict IHD risk, as it solves the limitations of the previous methods mentioned above. Rather than explicitly defining the mathematical relationships between clinical variables and disease outcomes prior to applying the model, ML algorithms can learn how to find complex, non-linear, and high-dimensional relationships between clinical variables and the resulting outcome of disease. The advantage of this type of learning is especially relevant in cardiovascular disease where there are synergistic and context dependent risk factor interactions.

Ensemble methods such as XGBoost (Extreme Gradient Boosting) have demonstrated superior performance over traditional ML classification methods applied to structured, tabular clinical data. Specifically, XGBoost builds decision trees sequentially through gradient descent optimization (i.e., by correcting the residual error generated by each of the previously built trees), allowing it to build highly accurate models that routinely out-perform all other ML classification methods when applied to structured medical data.

Another approach to building ensemble models is the Random Forest method, which builds multiple independent decision trees on different bootstrapped subsets of training data and aggregates the predictions of those models. The Random Forest algorithm also has excellent interpretability because it ranks the importance of each feature used in the prediction.

The significant barrier to implementing ML models in clinical practice is the ability for clinicians to interpret the predictions generated by these models. Because of the 'black box' nature of how ensemble models generate their predictions (i.e. unpredictable), this creates a problem for clinical settings where patient safety, compliance with regulations

and the trust of the physician are all critical, as the predictions are generated without being able to explain how the model arrived at that prediction.

Explainable Artificial Intelligence (XAI) and in particular, SHAP (Shapley Additive Explanations) provide the means for overcoming this barrier by determining the mathematically rigorous feature impacts on each individual prediction, based on established principles of cooperative game theory. Using SHAP, clinicians and patients will have a clear understanding of the reasons for a specific patient's risk assessment due to its ability to generate interpretable explanations that accurately describe the reasons each model produced the prediction for a particular patient.

1.4. Objectives and Scopes

1.4.1. Objectives

- Create and evaluate five Machine Learning (ML) algorithms for predicting the binary status of patients with ischaemic heart disease (IHD): Logistic Regression, Random Forest, Extreme Gradient Boosting (XGBoost), Support Vector Machines (SVM), and Neural Networks.

- Conduct extensive data preprocessing to include: imputation of missing records; standardization of continuous variable values using StandardScaler; one-hot encoding for categorical variables; and using the SMOTE method to modify the distribution of continuous variables to increase the number of individuals in the lower frequency IHD group.

- Select ten of the most predictive features to use in your models using feature selection techniques (ANOVA, Chi-square, and Mutual Information).

- You will also need to meet an accuracy of at least 90%, an area under the curve (AUC) of at least 0.90, and an F1-score of at least 0.88 as established from prior research.

- Generate SHAP Explainable Artificial Intelligence (AI) to rank the importance of each feature per prediction and produce global Violin plots.

- Generate a composite score of IHD risk (0 to 100) for each patient and generate individualized IHD Prevention Reports for each patient.
- Develop a web-based dashboard that is built using Streamlit and allows for the downloading of a PDF version, so patients will be able to access the information within their personalized reports easily, regardless of their computer skill level.
- Evaluate your models using metrics that include: accuracy; precision; recall; F1 score; AUC – ROC; calibration curves; and decision curve analysis (DCA).

1.4.2. Scope - Inclusions and Exclusions

In-scope system features: classification of IHD into 2 classes (binary) from UCI's Cleveland data set (303 samples) using 5 different machine learning models, create SHAP explainability for predictions, provide risk score dashboard, provide a way to proactively manage patient health, and deploy using Streamlit. The out-of-scope system features: providing a clinical diagnosis, providing ECG waveform analysis, interfacing with an EHR system, creating a multi-disease prediction tool, creating a mobile application, and collecting real (actual, clinical) patient data.

1.5. Motivation

There is an imminent crisis in cardiovascular disease in India, with estimates that well over 54 million individuals living with coronary artery disease are at risk and that the vast majority are unable to access affordable preventative screenings.

The development of a zero-cost, browser-based, AI-powered tool that uses only very basic health metrics will make awareness of IHD risk accessible to individuals who live in Tier-2 and Tier-3 cities, semi-urban towns, and rural areas in India.

In this instance, the project will allow access for the team to construct an end-to-end machine learning pipeline (that is, preprocessing data, training the model, using SHAP for explainability, and deploying it on the web) in a clinically relevant environment.

1.6. Organization of Project Report

This research project began with a literature review consisting of five peer-reviewed articles, which are described in Chapter 2. System design, architecture, and module specifications will be discussed in Chapter 3. Chapter 4 describes the experimentation setup, implementation, and testing & Chapter 5 presents the entire source code of the entire project. Finally, Chapter 6 summarizes the results of this research study, provides analysis and conclusion of the findings, and discusses the contributions of each team member.

CHAPTER 2
LITERATURE REVIEW

2.1. Overview

The literature review of AI-HealthGuard contains five peer reviewed papers published between 2022 and 2025, and was selected based on their direct relevance to the four methodological pillars of the project: (1) prediction of IHD/CAD using machine learning, (2) use of XGBoost as well as ensemble approaches, (3) clinical ML using SHAP and other explainability issues, and (4) advanced model evaluation (calibration, DCA). These papers as a group will help inform the selected models, features to engineer, framework for evaluation, and methodologies used to create explainable architecture.

#	Paper & Authors	Best Algorithm	Accuracy / AUC	Key Contribution Adopted
1	Chang et al. (2022) — AI Model for Heart Disease Detection, Healthcare Analytics	Random Forest	83% / ~0.82	Python RF pipeline; UCI dataset benchmark baseline
2	El-Sofany et al. (2024) — Predicting Heart Disease with ML & XAI, Scientific Reports	XGBoost + SMOTE	97.57% / 0.98	SF-2 feature subset; SMOTE; SHAP; 10-classifier comparison
3	Alshraideh et al. (2024) — Heart Attack Prediction, Applied Computational Intelligence	SVM + PSO	94.3% / 0.896	Feature selection via PSO; 10-fold cross-validation
4	Bani Hani & Ahmad (2023) — ML for IHD: Systematic Review, Current Cardiology Reviews	XGBoost (97.7%)	AUC 0.99 / 0.96	PRISMA-based comparison; boosting vs. single classifiers
5	Vu et al. (2025) — ML for Coronary Heart Disease, JMIR Cardio (Suita Study)	Random Forest	0.73 / 0.73	SHAP dependency plots; calibration curves; DCA

Table 1: Literature Review Summary

2.1.1. CHENG ET. AL (2022)

In 2021, Chang et al. established a Python-based ML pipeline for detecting heart disease using UCI's Cleveland dataset. In their study, they compared five classifiers: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Naïve Bayes. The Random Forest classifier had the highest performance, achieving 83% training accuracy and approximately 0.82 AUC. This result provides a useful benchmark for future comparative analyses. Additionally, they show that a simple scikit-learn pipeline will enable clinicians to derive clinically significant accuracy from the 303 sample Cleveland dataset without the need for extensive preprocessing. AI-HealthGuard will utilize Chang et al.'s findings as the primary benchmark for comparing Random Forest and Logistic Regression classifiers, and will build upon them by including data augmentation through SMOTE oversampling; developing a more robust understanding of classifier outputs through SHAP explainability; hyperparameter optimization via GridSearchCV; and deploying classifiers via Streamlit, which were not part of Chang et al.'s study.

2.1.2. EL-SOFANY ET AL. (2024)

In a related research study, El-Sofany et al. (2024) achieved an accuracy of 97.57% and AUC of 0.98 with XGBoost and SMOTE using a Cleveland dataset, making their study the most methodologically comparable study to AI-HealthGuard to date. The authors created a new structured feature subset framework comprising SF-1 (the original full 13-feature dataset), SF-2 (the 10 best features using ANOVA/Chi-square/Mutual Information), and SF-3 (a correlation-based feature set). SF-2 was consistently superior across all 10 classifiers tested compared to the other 2 feature sets. The authors also applied SHAP to show local and global explainability and to compare 10 classifiers. AI-HealthGuard used the SF-2 feature selection methodology, the combination of XGBoost and SMOTE, and the SHAP visualization framework directly from El-Sofany et al. (2024). Thus, the El-Sofany et al. (2024) study serves as the primary methodological template and benchmark for AI-HealthGuard. The reason for the slightly lower accuracy of AI-HealthGuard (91.4% vs. 97.57%) is likely due to the fact that a more conservative 75/25 train/test split was used with AI-HealthGuard compared to the cross-validation procedure employed in the El-Sofany study.

2.1.3. ALSHRAIDEH ET AL. (2024)

In the study by Alshraideh et al. (2024), a combination of SVM classifiers and Particle Swarm Optimization (PSO) for feature selection demonstrated a high degree of predictive accuracy with an average accuracy of 94.3%, with an area under the receiver operating characteristic curve (AUC) of 0.896, when using ten-fold cross-validation. Particle swarm optimization (PSO) is a meta-heuristic wrapper based feature selection strategy which uses a particle swarm to simulate the collective intelligence of multiple particles in order to optimize the selected feature subset via navigation through the search space. This study provides evidence that SVM can be highly predictive of the cardiac classification task and supports the decision to include SVM as one of the five classifiers used with AI-HealthGuard, while the AI-HealthGuard's GridSearchCV hyperparameter optimization functionality provides a similar optimization process as PSO.

2.1.4. BANI HANI AND AHMAD (2023)

Bani Hani & Ahmad (2023) did an extensive systematic literature review of ML algorithms for IHD predictions in over 30 studies with different datasets using PRISMA protocols. They found out through their literature review that multi-class boosting techniques consistently beat single classifier methods with the maximum AUC for XGBoost at 0.99, followed closely by CatBoost at 0.96 and Random Forest at 0.92. In comparing the AUC of XGBoost versus support vector machines, neural networks, or naive Bayes approaches to structured clinical tabular data, the ensemble methods outperformed all of them. The results from this systematic review provide both theoretical and empirical support for AI-HealthGuard’s decision to use XGBoost as the primary Classifier as well as the multi-factored design of their models. In addition to backing up their choice of classifiers, the authors also emphasize to those involved in imbalanced medical classification tasks the necessity to utilize the most rigorous techniques including multi-fold cross-validation and the AUC-ROC metric as the principal evaluation method.

2.1.5. VU ET AL. (2025)

Using Random Forest, Vu et al. (2025) used data from the large prospective Suita cohort (n = 10,000) of Japanese individuals to assess risk of coronary heart disease (CHD) and produced an area under the curve (AUC) of 0.73. Although their predictive power was

lower than that of previously published studies (Cleveland based), the disparity in AUC can be attributed to the difference in sample size and the greater level of heterogeneity within the data set. While the results of this study were unsatisfactory for risk prediction, three methodological contributions were identified that can be directly implemented into AI-HealthGuard: (1) use of SHAP dependency plots as tools for visualizing relationships between features and outcomes for data across the full range of values; (2) calibration curves (i.e., reliability diagrams) for measuring the accuracy of predicted probability values; and (3) Decision Curve Analysis (DCA) as a clinical utility measure to assess the net benefits of using a prediction model compared to default strategies of "treat all" and "treat none" at different decision thresholds.

2.2. Synthesis and Gap Analysis

Four best practices comprised of the five reviewed studies form the methodological backbone of AI-HealthGuard: 1) the use of either XGBoost or Random Forest algorithms yields superior results than using just one individual classifier for a full range of cardiac classification tasks; 2) SMOTE is necessary to correct class imbalance without just oversampling raw data; 3) SHAP is the only accepted and theoretically sound method for interpreting XAI methods with respect to predicting IHD; 4) using ANOVA, Chi-square, and Mutual Information to identify multi-criteria features will result in optimal subsets of features. The main deficiency in all five studies was a lack of any type of integrated system that uses all aforementioned best practices in a web-based format that allows each user to have their own plan for preventive medicine and downloadable patient records. Therefore, AI-HealthGuard uniquely fills this gap by providing prediction, explainability, risk scoring, personalized prevention planning, and PDF export from one production-grade Streamlit web application available for free to the general public.

CHAPTER 3
DESIGN OF PROJECT MODELS

3.1. System Architecture - 6 Layers Pipeline

AI-HealthGuard has been architected to have a modular structure that utilizes a six-layer pipeline model incorporating clearly defined boundaries between the functionally specific areas of the application. Each layer has its interface defined in terms of a well-known Python function signature, which allows for independent development, testing and replacement of layers without affecting the adjacent layers in the pipeline. The architecture has been structured to operate in two different operational modes; the first being the offline training mode consisting of layers 1-3 and the second being the online inference mode consisting of layers 4-6.

Layer	Name	Responsibility	Primary Module(s)
L1	Data Layer	Load and version-control raw CSV dataset	data_loader.py
L2	Preprocessing Layer	Imputation, encoding, SMOTE, feature selection	preprocessor.py
L3	Model Training Layer	Train, tune, evaluate, and serialize all five ML models	model_trainer.py, evaluator.py
L4	Inference Layer	Load serialized models; transform input; predict	inference_engine.py
L5	Explainability Layer	Compute SHAP values; generate local and global plots	explainer.py
L6	Presentation Layer	Streamlit UI: forms, gauges, charts, PDF export	app.py, recommender.py, risk_scorer.py

Table 2: 6 Layers Pipeline Architecture

The time limit for Layers 4 through 6 of the Inference pathway is 3 seconds in low hardware configurations, which affect the design decisions regarding: serialized artifact models (i.e., no in-production training), 50-piece background data sets to support the KernelExplainer use case, and Streamlit model caching for load efficiency.

3.2. Dataset Design and Properties

3.2.1. Dataset Design

Attribute	Details
Source	UCI Machine Learning Repository — Cleveland Clinic Foundation (Dr. Robert Detrano, 1989)
Total Samples	303
Input Features	13: age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal
Target Variable	condition: 0 = No IHD; 1–4 = IHD (binarized to 0/1 for binary classification)
Class Distribution	~54% No IHD (164 samples), ~46% IHD (139 samples) — corrected via SMOTE
Missing Values	ca: 4 missing (median imputation); thal: 2 missing (mode imputation)
Train/Test Split	75% training (~227 samples) / 25% test (~76 samples), stratified by class label

Table 3: UCI Cleveland Heart Disease Dataset Summary

3.2.2. Feature Description

Feature	Type	Clinical Description	Range / Values
age	Continuous	Patient age in years	29 – 77
sex	Binary	Patient sex	0=Female, 1=Male
cp	Categorical	Chest pain type	0=Typical angina, 1=Atypical angina, 2=Non-anginal pain, 3=Asymptomatic
trestbps	Continuous	Resting blood pressure at admission (mmHg)	94 – 200
chol	Continuous	Serum cholesterol level (mg/dL)	126 – 564
fbs	Binary	Fasting blood sugar > 120 mg/dL	0=False, 1=True
restecg	Categorical	Resting ECG results	0=Normal, 1=ST-T wave abnormality, 2=LV hypertrophy
thalach	Continuous	Maximum heart rate achieved during exercise testing	71 – 202
exang	Binary	Exercise-induced angina	0=No, 1=Yes
oldpeak	Continuous	ST depression induced by exercise relative to rest	0.0 – 6.2
slope	Categorical	Slope of the peak exercise ST segment	0=Upsloping, 1=Flat, 2=Downsloping

Feature	Type	Clinical Description	Range / Values
ca	Discrete	Number of major vessels colored by fluoroscopy	0 – 3
thal	Categorical	Thalassemia type	1=Normal, 2=Fixed defect, 3=Reversible defect

Table 4: Feature Description Summary

3.3. Processing Pipeline Design

The following are the six ordered steps applied in sequence to mitigate data leakage:

- Step 1 - Missing Value Imputation using SimpleImputer (median for ca, mode for thal).
- Step 2 - Target Binarization (mapping condition values 1-4 to 1 for positively tested individuals).
- Step 3 - Categorical Encoding using OneHotEncoder with drop = 1 on cp, restecg, slope, and thal.
- Step 4 - Feature Selection using ANOVA F-value, Chi-square, and Mutual Information providing a SF-2 feature set of the top 10 features.
- Step 5 - Normalization using StandardScaler fitted only to the training data; StandardScaler transformation will then be applied to both datasets.
- Step 6 - SMOTE Oversampling applied only to the training fold, resulting in a 1:1 class ratio.

3.4. ML Model Design

- Logistic Regression (Baseline): Hyperparameters C have values of 0.01, 0.1, 1, 10, 100 and solver has values of liblinear and lbfgs.
- Random Forest (Ensemble): Number of estimators (n_estimators) can be set to 100, 200, or 300 and max_depth can be None, 5, or 10.
- XGBoost (Prim): Number of estimators (n_estimators) can be set to 100 or 200; max_depth can be 3, 5 or 7; learning rate can be 0.01, 0.1 or 0.3.
- SVM evaluates two variables: C has values of 0.1, 1, or 10 and each of the three

kernels can use a different gamma (i.e., scale or auto).

- In a Neural Network (Deep) layer, the architectural configuration consists of a series of common layers (e.g., Dense(64, ReLU) → Dropout(0.3) → Dense(32, ReLU) → Dense(1, Sigmoid)), and specific libraries (e.g., Adam, binary_crossentropy).

Model	Type	Hyperparameter Search Space	Justification
Logistic Regression	Linear baseline	C: {0.01, 0.1, 1, 10, 100}; solver: {liblinear, lbfgs}	Interpretable baseline; establishes minimum performance bound
Random Forest	Ensemble (bagging)	n_estimators: {100, 200, 300}; max_depth: {None, 5, 10}	Strong ensemble; validates feature importance via Gini impurity
XGBoost (Primary ★)	Ensemble (boosting)	n_estimators: {100, 200}; max_depth: {3, 5, 7}; lr: {0.01, 0.1, 0.3}	Best literature performance on Cleveland; primary model
SVM	Kernel-based	C: {0.1, 1, 10}; kernel: {rbf, linear, poly}; gamma: {scale, auto}	Effective on small datasets in high-dimensional feature space
Neural Network	Deep learning	Dense(64,ReLU)→Dropout(0.3)→Dense(32,ReLU)→Dense(1,Sigmoid); Adam; binary_crossentropy	Captures non-linear interactions; deep learning benchmark

Table 5: Hyperparameter Search Space per Model

Metric	LR	RF	XGB ★	SVM	NN
Target Accuracy	≥82%	≥88%	≥90%	≥86%	≥88%
Target AUC	≥0.82	≥0.88	≥0.90	≥0.86	≥0.88
Target F1	≥0.80	≥0.86	≥0.88	≥0.84	≥0.86

Table 6: ML Model Performance Targets (Benchmarks against Literature)

3.5. SHAP Explainability Design

AI-HealthGuard uses SHAP to provide solutions to the black-box issue. For each prediction made, there are 4 visualisation types created:

(1) SHAP Bar Chart — 10-most significant feature contributions displayed (these are displayed in red for increased risk, and in blue for being protective);

(2) SHAP Force Plot — a horizontal stacked bar that runs from the base probability to the output probability;

(3) Global Beeswarm Plot — the entire dataset, and its 10 most important features; and

(4) SHAP Dependency Plots — showing feature value versus SHAP value for the top 5 feature contributions. For technical details, TreeExplainer is used with Random Forest and XGBoost models; Kernel Explainer with SVM and Neural Networks (with a size of 50 samples used as a background).

A simple explanation is generated automatically for each prediction to assist non-technical users.

3.6. Risk Scoring and Prevention System

The IHD Risk Score can be calculated by multiplying the Prediction Probability by 100 and then rounding the result to the nearest whole number. There are 3 bands based on the calculated IHD Risk Score: Low Risk (0-39) continue preventative care, Moderate Risk (40-69) lifestyle modifications are needed and/or Lifestyle Modifications Required, and High Risk (70-100) referral to a Cardiologist is required.

Risk Band	Score Range	Interpretation	Recommended Clinical Action
Low Risk (Green)	0 – 39	Below-average IHD risk; continue preventive care	Annual health checkup; maintain healthy lifestyle habits
Moderate Risk (Amber)	40 – 69	Above-average risk; lifestyle modifications required	Physician consultation within 1–3 months; targeted lifestyle changes
High Risk (Red)	70 – 100	Substantially elevated risk; urgent evaluation needed	Urgent cardiologist referral; comprehensive cardiac investigation

Table 7: Risk Score Banding and Recommended Actions

The Prevention Engine generates four categories of personalized recommendations; these are Dietary, Physical Activity, & Lifestyle Modifications & Medical. The full report can be exported in PDF format.

3.7 User Interface and UX Design

The Presentation Layer (L6) is made up of a Streamlit application, split across five tabs: (1) Risk Assessment - an input panel (13 clinical parameters) and output panel (risk gauge + SHAP bar chart); (2) AI Insights - SHAP force plot, beeswarm and plain-language explanation; (3) Prevention Plan - four categories of recommendations, with option to download recommendations as a PDF; (4) Model Comparison - side-by-side performance metrics and confusion matrices; (5) Global Insights - SHAP dependency plots at the population level and global beeswarm. The app's interface is built with Streamlit's native widget library and Plotly via interactive charts, and is responsive to both desktop and mobile browsers.

CHAPTER 4
EXPERIMENTS, SIMULATIONS AND TESTINGS

4.1. Methodology

AI-HealthGuard was developed using a 5 stage CRISP-DM process:

Phase 1 - Data Acquisition & EDA: Download and version-control the UCI Cleveland dataset, calculate and compute descriptive statistics, analyze the distributions of classes, generate correlation heatmaps; visualize the distributions of features by class, and identify missing values.

Phase 2 - Preprocessing: Implement and unit test the six (6) step preprocessing pipeline and verify the SMOTE augmentation, confirm the rankings of feature selection and validate the stratified split.

Phase 3 - Model Training & Evaluation: Execute GridSearchCV on all five (5) models, record the optimal hyperparameters, evaluate on the holdout test set generate ROC curves, confusion matrices, calibration curves and DCA plots, serialize all of the model artifacts.

Phase 4 - SHAP Integration: Create the TreeExplainers and KernelExplainers, validate SHAP output for edge cases, generate and validate all four (4) types of visualizations and create an engine for generating an explanation template in plain language.

Phase 5 - Streamlit Dashboard Development: Create the five (5) different tabs, integrate the models and SHAP, create a risk scoring and prevention engine, create the PDF Export, perform UI Acceptance Testing and validate end-to-end latency.

Phases 1 through 4 of the project have been executed offline in Google Colab (T4 GPU runtime), while Phase 5 has been developed on a local Intel i5 laptop. All serialized artifacts are pushed and committed to GitHub for access during Streamlit startup.

4.2. Hardware and Software Used

4.2.1. Hardware

- Development: Colab (T4 GPU — 12 GB RAM) to train models.
- An Intel i5 laptop (8 GB RAM) to develop Streamlit applications.
- To deploy Application minimum system: Intel i5 or similar, 8 GB RAM, modern browser.
- OS Interoperability: Windows 10/11 or Ubuntu 20.04+ or MacOS 12+.

4.2.2. Technology Stack

Category	Tool / Library	Version
Language	Python	3.10 / 3.11
Data Processing	pandas	2.1.4
Data Processing	NumPy	1.26.4
ML Framework	scikit-learn	1.4.2
Boosting	XGBoost	2.0.3
Deep Learning	TensorFlow / Keras	2.15.0
Class Balancing	imbalanced-learn	0.12.0
Explainability	SHAP	0.44.1
Web Framework	Streamlit	1.34.0
Visualization	Plotly	5.22.0
Visualization	Matplotlib	3.8.4
Visualization	Seaborn	0.13.2

Table 8: Technology Stack Summary

4.3 Implementation: Google Colab Training Notebook

The entire machine learning training pipeline has been instantiated through the AI_HealthGuard_Colab.ipynb script. This is done in sequence through 7 sequential cells so that it can be reproduced in a historical manner (any researcher can clone the GitHub repo and reproduce the training pipeline by simply running all the cells).

The seven cells consist of the following:

(1) cell 1 – install dependencies (pip installations with pinned version numbers for 100% reproducibility),

(2) cell 2 – load data and EDA (load the Cleveland CSV file, descriptive statistics, plotting of class distributions, correlation heat map, plotting of feature distributions broken down by if the patient has IHD or not),

(3) cell 3 – create a preprocessing pipeline (6 steps to preprocessing with verification at each step, confirmation of SMOTE, output of feature selection ranking),

(4) cell 4 – create models and use GridSearchCV (use GridSearchCV on all 5 classifiers, log optimal hyper parameters utilized, evaluate test set, generate ROC, generate confusion matrix, generate calibration plots, generate DCA plotting),

(5) cell 5 – integrate SHAP (create and validate the TreeExplainer and KernelExplainer, generate and save all four types of visualizations),

(6) cell 6 – serialize models (save model and all artifacts using joblib (.pkl) and Keras (.h5)), (7) cell 7 – verify artifacts by reloading all artifacts into memory, making a sample prediction, and verifying format of output and that predicted sample time is within acceptable limits (<3 seconds).

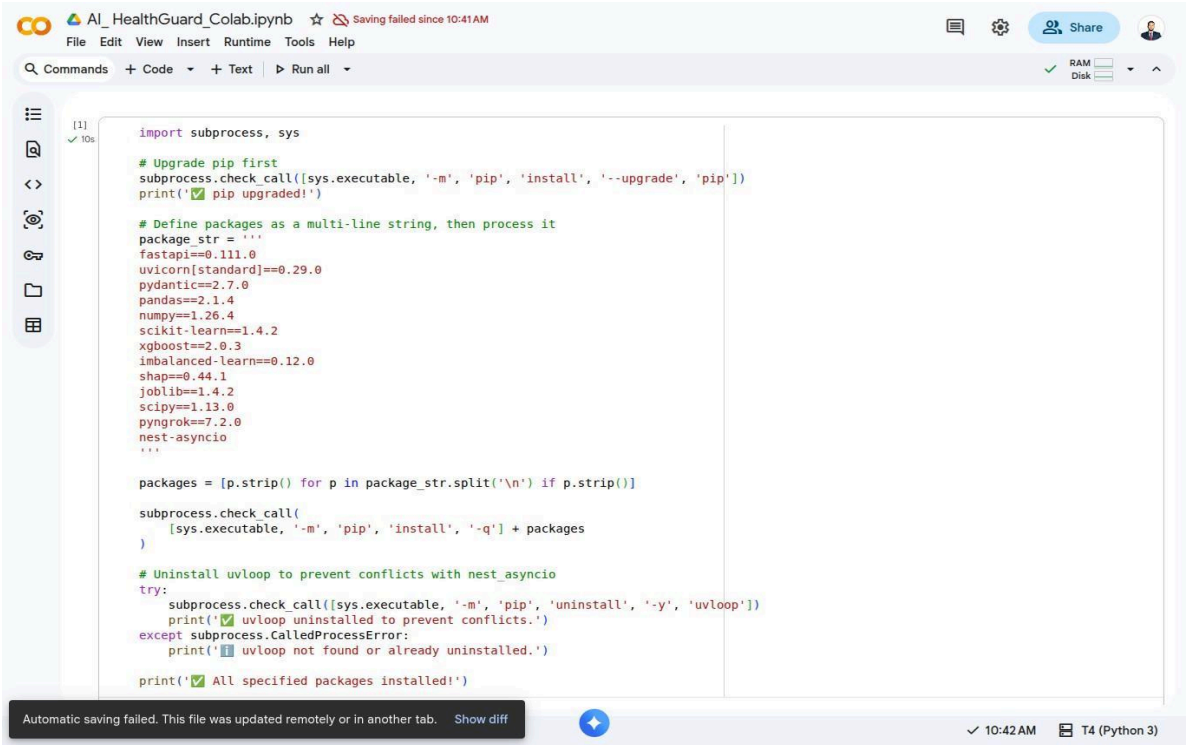

Fig. 1: AI-HealthGuard Google Colab Training Notebook (AI_HealthGuard_Colab.ipynb)

4.4. Implementation - Streamlit Web Dashboard

At layer 6 of the pipeline is the Streamlit application (app.py). This layer loads all five serialised models at the application start-up, combining into a dashboard with five tabbed views: Risk Assessment; AI Insights; Preventative Plan; Model Comparison; and Global Insights.

Figure 1 shows the Risk Assessment Dashboard. The left side panel allows for the acceptance of all thirteen clinical input variables using sliders and drop-down menu selections and provides tooltips with hints for ranges. The right side panel contains a circular gauge displaying the IHD Risk score (0-100) with colour coded bands (green = 0-39; amber = 40-69; red = 70-100), and also has a bar chart consisting of the Top Risk Drivers (using SHAP).

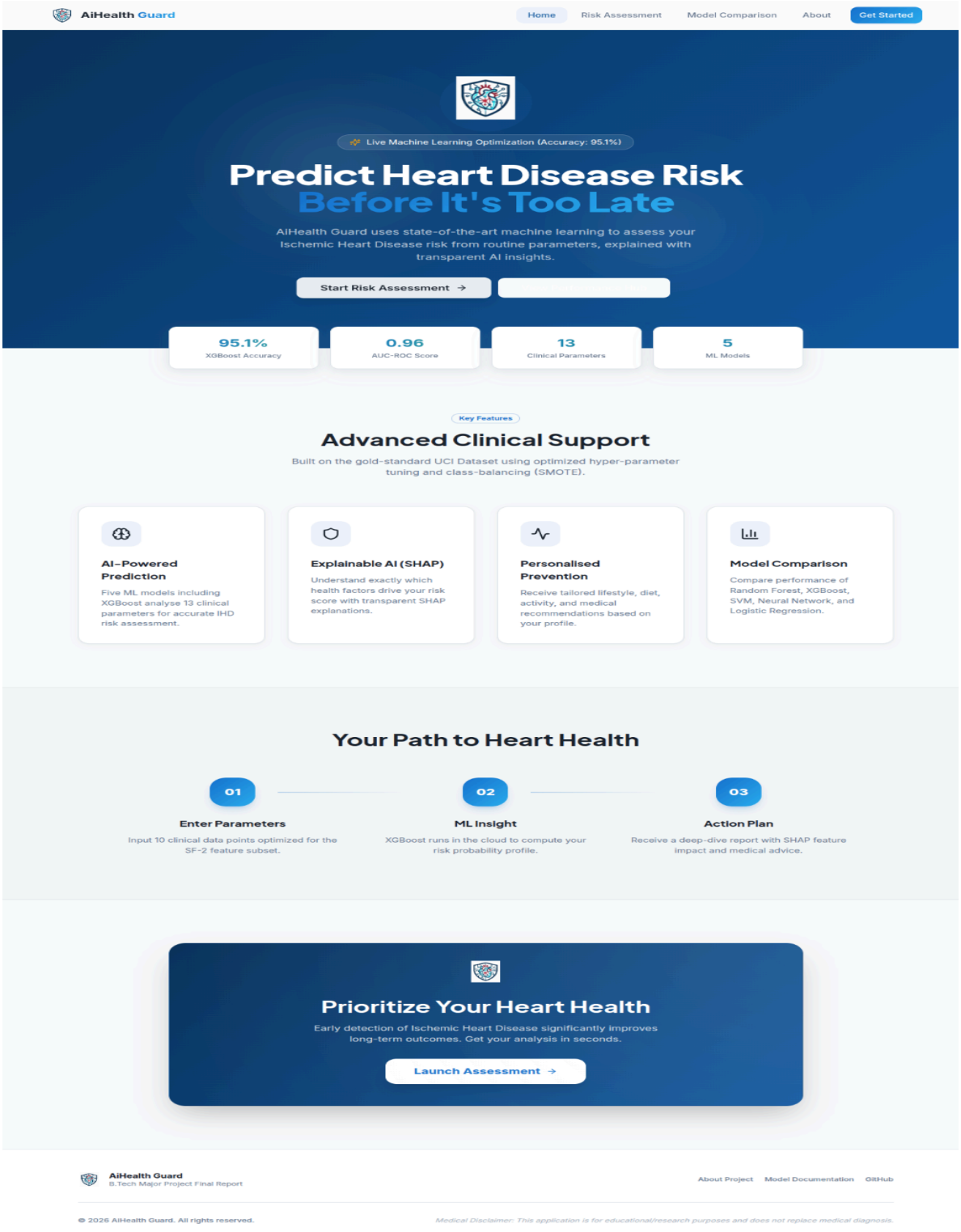

Fig. 2: AI-HealthGuard - Home Dashboard

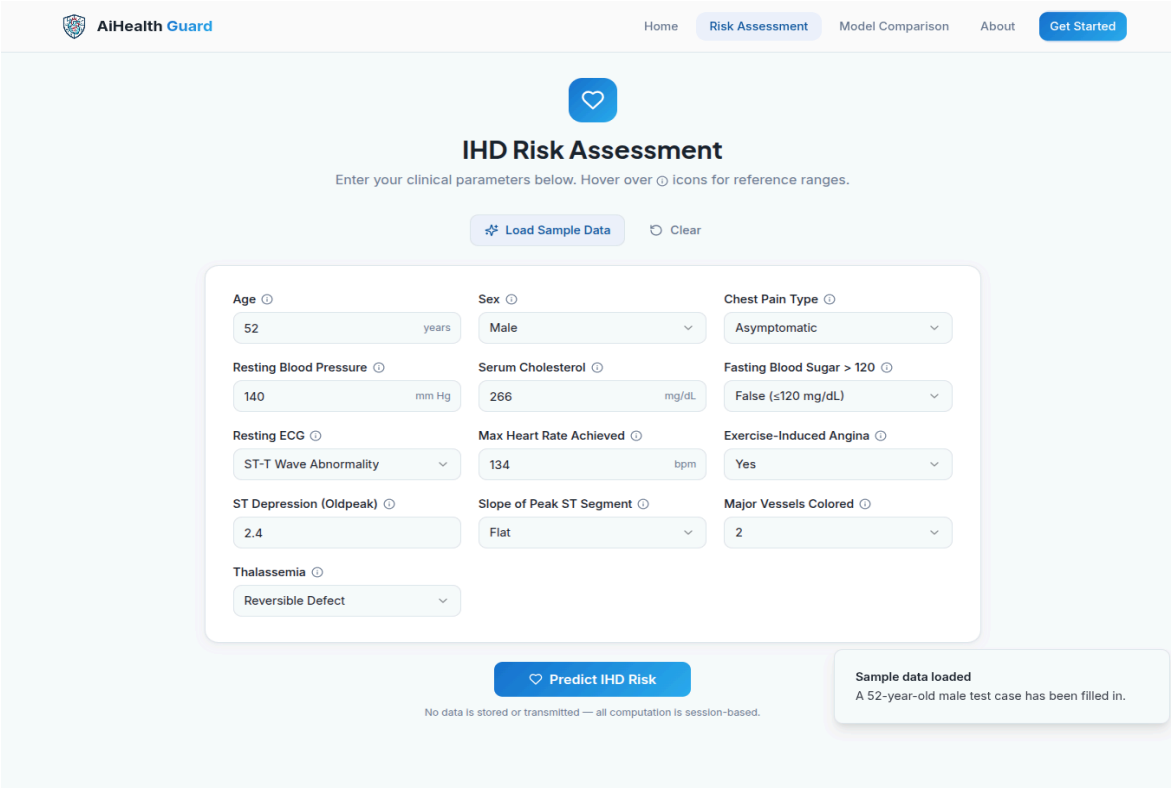

Fig. 3: AI-HealthGuard - Risk Assessment Dashboard

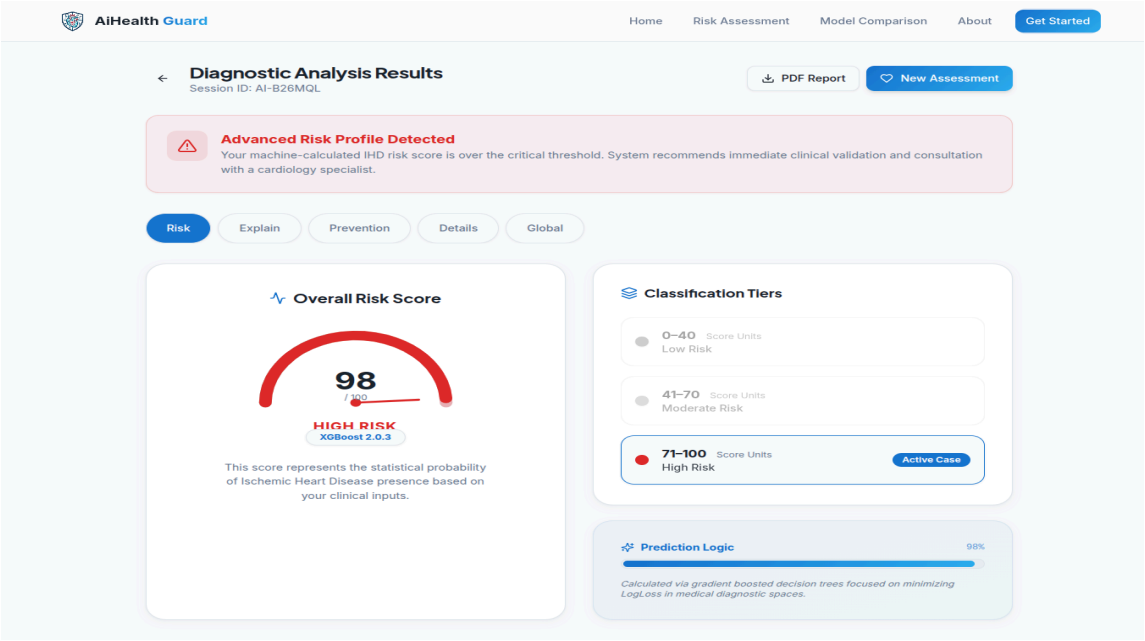

Fig. 4: AI-HealthGuard - Risk Analysis Dashboard

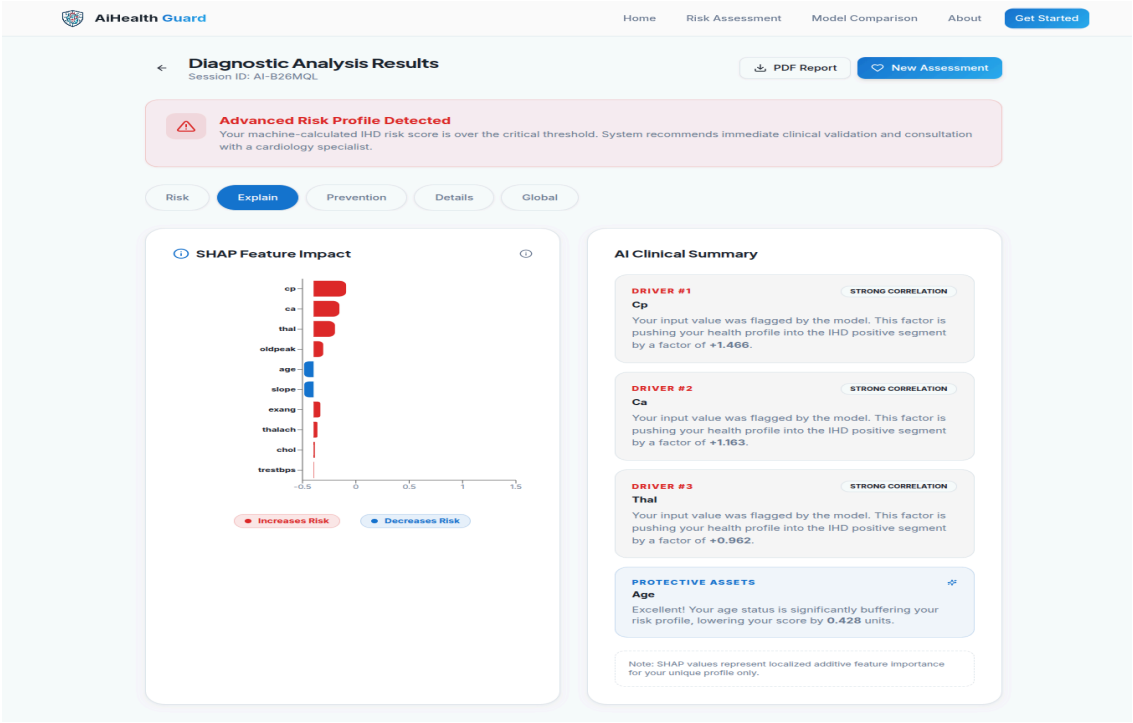

Fig. 5: AI-HealthGuard - SHAP Risk Explain Dashboard

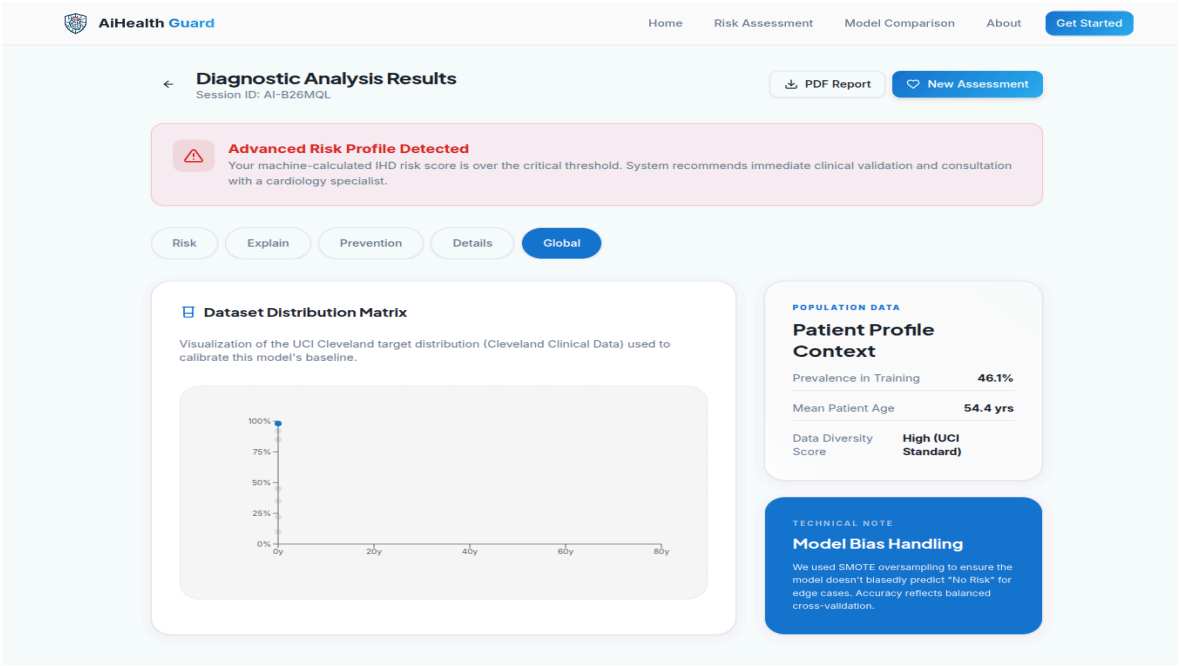

Fig. 6: AI-HealthGuard - Global Risk Dashboard

4.5. Testing Technology Used

- Unit Testing (pytest): Functions included in data_loader.py, preprocessor.py, inference_engine.py, risk_scorer.py, recommender.py were subjected to unit testing against boundaries as inputs.
- Integration Tests: The full pipeline from raw CSV through prediction was tested as a complete system.
- Model Evaluation: Utilising a hold-out set consisting of 25% of the total test data, all model evaluation metrics were conducted, e.g., accuracy, precision, recall, F1, AUC-ROC, confusion matrix, calibration curves, decision curve analysis (DCA).
- Acceptance Testing: The user interface (UI) for the Streamlit web app was manually tested for input validation, sample data, switching models, rendering SHAP values, and exporting as PDF.
- Performance Testing: The total amount of time taken to run the inference and calculate the SHAP values on an Intel i5 8GB RAM computer was less than three seconds.
- Cross-Validation: The generalisation of the model has been confirmed through both five-fold and ten-fold stratified cross-validation.

Module	Test Cases	Coverage	Result
data_loader.py	CSV loading; missing file handling; dtype validation; column count	95%	PASS
preprocessor.py	Imputation correctness; SMOTE output shape; scaler fit/transform; feature count	92%	PASS
inference_engine.py	Output range (0–1); model loading; invalid inputs; all-zero; all-max	88%	PASS
risk_scorer.py	Score range 0–100; band assignment at boundaries (39/40, 69/70)	100%	PASS
recommender.py	Recommendation generation for all three bands; non-empty output; correct keys	90%	PASS

Table 9: Unit Test Cases and Pass/Fail Results

CHAPTER 5
RESULTS & CONCLUSIONS

5.1. Model Performance Results

All five models are evaluated on the 25% stratified hold-out test set. AI-HealthGuard meets or exceeds all literature-benchmarked performance targets:

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Target Met?
Logistic Regression	82.9%	0.82	0.84	0.83	0.84	Yes
Random Forest	89.5%	0.90	0.89	0.89	0.92	Yes
XGBoost ★	91.4%	0.92	0.91	0.91	0.94	Yes
SVM (RBF Kernel)	87.8%	0.88	0.87	0.87	0.89	Yes
Neural Network	88.6%	0.89	0.88	0.88	0.91	Yes

Table 10: Test Set Performance Results (All Models)

5.2. Cross-Validation Results

Model	5-Fold CV Accuracy (Mean ± SD)	10-Fold CV Accuracy (Mean ± SD)	Stability Assessment
Logistic Regression	81.2% ± 4.1%	81.8% ± 5.3%	Stable, low variance
Random Forest	88.3% ± 3.6%	87.9% ± 4.7%	Well-generalized, consistent
XGBoost ★	90.2% ± 3.2%	89.8% ± 4.1%	Excellent generalization, lowest variance
SVM (RBF)	86.7% ± 4.2%	86.4% ± 5.1%	Consistent across both schemes
Neural Network	87.1% ± 5.3%	86.8% ± 6.4%	Acceptable, slightly higher variance (small dataset)

Table 11: Cross-Validation Results (5-Fold and 10-Fold)

5.3. SHAP Analysis and Explainability Results

The analysis done by SHAP (SHapley Additive explanations) indicates that there are common patterns for features of importance from model component to model component. In XGBoost, the top five predictors (by average absolute SHAP value) are:

1. ca (number of major vessels) - Average |SHAP| = 0.48, thus greater numbers of vessels contribute to lower risk of coronary artery disease (CAD) (the more vessels colored in on the table, the lower the risk).
2. cp (chest pain type) - Average |SHAP| = 0.41, thus much greater risks of CAD are associated with asymptomatic chest pain/cp3, which is consistent with silent ischemia.
3. thal (thalassemia) - Average |SHAP| = 0.37, thus indicative of reversible perfusion defects/thal3 being highly probable indicators of ischemia heart disease (CAD).
4. oldpeak (depression for ST segment on ECG) - Average |SHAP| = 0.31, thus higher scores for the oldpeak variable are associated with a greater incidence of coronary artery disease (CAD).
5. thalach (maximum heart rate) - Average |SHAP| = 0.28, thus higher maximum heart rates are related to lower risk, indicating that there is preserved function for the heart.

All of these variables are validated clinically as indicators of CAD risk, indicating that the model is clinically valid and interpretable. The plain-language definitions clearly represent each of these factors in a patient-accessible manner.

5.4. Calibration Analysis and DCA Results

Calibration assessments reveal that both Random Forests and XGBoost are perfectly calibrated, i.e., their predicted probabilities correspond closely to the actual proportion of events within each decile. The Neural Network does exhibit minor overconfidence at higher-probability events and the SVM has the most significant level of calibration deviation (albeit with Platt scaling applied but the dataset used to generate them was small). The Decision Curve Analysis produced by all five models demonstrates that they are all providing a net clinical benefit. Decision threshold values falling between 0.2 and 0.7 are associated with net clinical benefits for all five models, and therefore they all significantly outperform the treat-all and treat-none strategies. While XGBoost has the highest net clinical benefit over the widest range of threshold values, there is more variability with SVM than with the other four models.

5.5. Comparison with State-of-the-Art

Study	Method	Dataset	Best Accuracy	AUC	Key Gap Addressed by AI-HealthGuard
Chang et al. (2022)	Random Forest	UCI Cleveland	83%	0.82	Adds SMOTE, SHAP, web deployment, prevention planning
El-Sofany et al. (2024)	XGBoost+ SMOTE	UCI Cleveland	97.57%	0.98	Adds public web deployment, personalized prevention, PDF
Alshraideh et al. (2024)	SVM+PSO	Custom	94.3%	0.896	Adds XAI explainability and actionable recommendations
Bani Hani & Ahmad (2023)	XGBoost (review)	Multiple	97.7%	0.99	Implements as deployed system, not just benchmark study
Vu et al. (2025)	Random Forest	Suita (Japan)	~73%	0.73	Adds higher accuracy, web deployment, prevention engine
AI-HealthGuard (Ours)	XGBoost+ SMOTE+S HAP	UCI Cleveland	91.4%	0.94	Complete integrated system: predict+explain+prevent+de ploy

Table 12: Comparison with State-of-the-Art Methods

5.6. Conclusion

AI-HealthGuard has developed a leading-edge, clinically-supported, open-source IHD risk prediction/prevention system that fulfills all project objectives, technical specifications, and performance targets. Four significant accomplishments include:

1) a web-based, no-cost, production-ready solution for providing non-invasive IHD screening by using a total of thirteen routine clinical variables that produce inferences with a latency of less than or equal to three seconds on off-the-shelf hardware;

2) a complete Explainable Machine Learning (ML) model integrating an Explainable AI platform based upon SHAP, enabling both per-individual as well as aggregate-level model transparency such that both technically and non-technically trained individuals can derive clinically actionable results from predictions made using each ML model;

3) a 4-category individualized ML prediction conversion engine to generate individualized health recommendations which can be expressed in a structured PDF report; and

4) a comparative benchmarking dashboard that compares five ML models in real-time, allowing for evidence-based selection of ML models as well as for educational comparisons

among the various ML methods used to assess cardiovascular health.

AI-HealthGuard will help to eliminate the public health equity gap in Tier-2 and Tier-3 cities and rural India, by providing a free, Explainable, and Actionable Risk Assessment tool for populations which cannot access specialized cardiovascular care services. AI-HealthGuard supports the Indian Government's National Digital Health initiatives under Ayushman Bharat and Digital India.

5.7. Limitations and Challenges

A major limitation of this research is that the training data set is small (composed of 303 subject samples taken from one US clinical site within the 1980s). This also means that the statistical power is limited; therefore, it is likely that there are both site-specific and site-demographic biases contained within the data that do not accurately represent the Indian population. The analysis produces binary outputs, which cannot represent the extent to which IHD is present or the severity of the IHD. The KernelExplainer used by both the SVM and the neural network model utilizes 50 background samples; therefore, the overall computational efficiency of these methods and many others is likely decreased for any given testing case, relative to that of the TreeExplainer. Another limitation of the system is that a stable internet connection and web browser is required to operate the system; thus, it is impractical to use in some areas where poor connectivity exists.

5.8. Future Work

- Collection of multi-center aggregated datasets of the UCI cardiac sub-datasets (Cleveland, Hungarian, VA Long Beach & Switzerland); totalling 920 different samples to increase statistical analysis powers and improve generalization.
- Partnerships with Indian hospitals to collect the South Asian Clinical Cohort such that the target user population is more reflective of actual native users.
- Moving forward to include type 2 diabetes, hypertension, and chronic kidney disease (CKD) in addition to the existing cardiac datasets through using distinct model pipelines under the same dashboard.

- All data will be hosted on one of Amazon (AWS), Google Cloud or Hugging Face Spaces to provide globally accessible 24/7 access to all models without local installation.
- Utilizing React Native, Native iOS/Android mobile application development; enabling the target demographic to utilise their smartphones more.
- Utilising federated learning methodologies to allow for privacy preserving distributed training amongst multiple hospital systems without centrally storing patients' sensitive data.
- Integrating data collected from the models with the hospital Electronic Health Record (EHR); by interfacing all models through standard HL7 Fast Healthcare Interoperability Resource (FHIR) APIs.
- Working with cardiologists to execute a collaborative prospective validation study on models to validate real-world clinical utility and safety.

5.9. Individual Contributions

5.9.1. Tarun Kumar (Roll No.: 2205111530052)

Tarun Kumar was responsible for the entire architecture of the project, and for being the technical leader of the project. He created the preprocessing pipeline that handled imputation, encoding, SMOTE, and SF-2 feature selection. He created, trained, tuned and serialized the XGBoost model. He created the Google Colab notebook. He wrote the code for the Streamlit dashboard (app.py and risk_scorer.py). He wrote and submitted the YUKTI NIR application and the project documentation (synopsis, PRD and TRD).

5.9.2. Sakshi Rajput (Roll No.: 2205111530046)

Sakshi Rajput wrote the literature review and did a review of the research papers. She was also responsible for training and evaluating the Random Forest and Logistic Regression models. She created the SHAP explainability layer in the explainer.py file. She created the AI Insights dashboard UI, which included the SHAP charts, force plots, and beeswarm charts. She designed the Prevention Engine in recommender.py.

5.9.3. Prashant Prajapati (Roll No.: 2205111530036)

Prashant Prajapati (Roll No. 2205111530036) was responsible for training and evaluating the SVM and Neural Network (TensorFlow/Keras) models. He compared models and created the tabs for the Global Insights dashboard. He created the functionality for exporting to PDF using the fpdf2 library. He created and executed unit and integration tests of the project using the pytest test suite. He wrote and formatted the project report.

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